



























































# How to measure $S_z(B^*)$ ?

★ Measure  $S_z(\gamma)$

$$B^* \rightarrow B + \gamma$$

Spin projection  
along the  $z$ -axis

$$1 = 0 + 1$$

|  | $S_z(B^*)$ | $S_z(\gamma)$ | $S_z(\nu)$     |
|--|------------|---------------|----------------|
|  | +1         | +1            | $-\frac{1}{2}$ |
|  | -1         | -1            | $\frac{1}{2}$  |
|  |            |               |                |

Photon helicity  
tags neutrino helicity

$$S_z(B^*) = S_z(\gamma)$$